The Force Table

Answer key

Conventions used throughout

• **360° scale:** 0° points right (+x), 90° up (+y), and angles increase counterclockwise, like the force table.

• **Components:** $F_x = F \cos \theta$ and $F_y = F \sin \theta$, using the 360° angle θ. The signs come out automatically (for example, cos 210° is negative, so that force points left).

• **Equilibrant** F_E: the one extra force that balances the others, so the ring doesn't move. With it included, $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

• **Final answers:** magnitude and 360° direction, written like the worksheet, for example 3.05 N [26.4°]. Full precision is kept in every step; only final answers are rounded.

FORCE TABLE ANALYSIS

Instructions for every set

1. Draw a Free-Body diagram of the three forces given above.
2. Calculate the magnitude of each component for the given forces.
3. Determine the components of the equilibrant using the net force in both the horizontal and vertical direction.
4. Determine the direction of the equilibrant using the 360° scale so it will be easy to test on the force table.
5. Test your answer at the link above.

SET 1

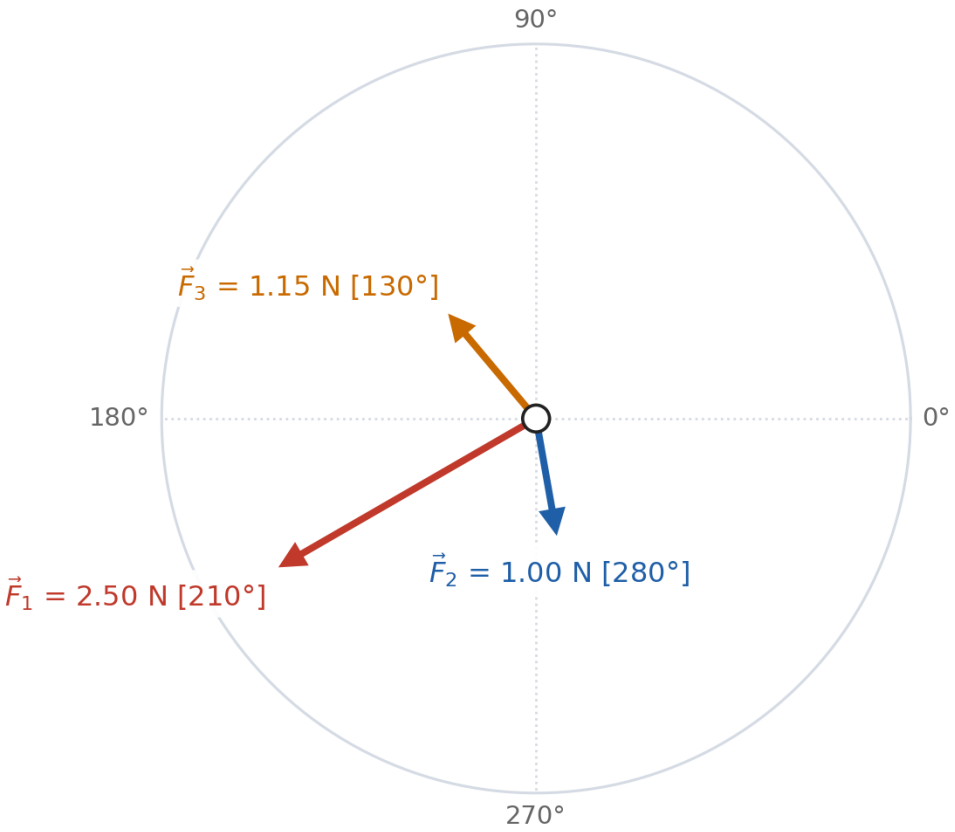
Forces to test

$F_1 = 2.50$ N [210°]
$F_2 = 1.00$ N [280°]
$F_3 = 1.15$ N [130°]

1. Draw a Free-Body diagram of the three forces given above.

Draw the ring as a dot. Each string pulls away from the ring at its angle on the 360° scale. Arrow lengths are proportional to the force sizes.

FBD of the ring (drawn to scale)



2. Calculate the magnitude of each component for the given forces.

Use the 360° angle of each force directly. A negative x component points left; a negative y component points down.

Formulas: $F_x = F\cos\theta$ $F_y = F\sin\theta$

$F_{1x} = (2.50)\cos 210° = -2.1651$ N

$F_{1y} = (2.50)\sin 210° = -1.2500$ N

$F_{2x} = (1.00)\cos 280° = 0.1736$ N

$F_{2y} = (1.00)\sin 280° = -0.9848$ N

$F_{3x} = (1.15)\cos 130° = -0.7392$ N

$F_{3y} = (1.15)\sin 130° = 0.8810$ N

Force	x component (N)	y component (N)
F_1 = 2.50 N [210°]	−2.1651	−1.2500
F_2 = 1.00 N [280°]	0.1736	−0.9848
F_3 = 1.15 N [130°]	−0.7392	0.8810
Sum	**−2.7306**	**−1.3539**

3. Determine the components of the equilibrant using the net force in both the horizontal and vertical direction.

With the equilibrant added, the ring is in equilibrium, so each direction adds to zero:

Formula: $\Sigma F_x = 0$

Write out the x forces: $F_{1x} + F_{2x} + F_{3x} + F_{Ex} = 0$

Put in the sum: $-2.7306 + F_{Ex} = 0$

Add 2.7306 to both sides: $F_{Ex} = 2.7306 \text{ N}$

Formula: $\Sigma F_y = 0$

Write out the y forces: $F_{1y} + F_{2y} + F_{3y} + F_{Ey} = 0$

Put in the sum: $-1.3539 + F_{Ey} = 0$

Add 1.3539 to both sides: $F_{Ey} = 1.3539 \text{ N}$

The equilibrant's components are the opposites of the sums.

4. Determine the direction of the equilibrant using the 360° scale so it will be easy to test on the force table.

Step 1 – Size of the equilibrant

F_{Ex} and F_{Ey} are at right angles, so use the Pythagorean theorem:

Formula: $F_E = \sqrt{F_{Ex}^2 + F_{Ey}^2}$

$F_E = \sqrt{(2.7306)^2 + (1.3539)^2} = 3.0478 \text{ N}$

Step 2 – Angle α from the x-axis

α is the angle between F_E and the x-axis. The y part is opposite α and the x part is next to it (use sizes, no signs):

$$\text{Formula: } \tan\alpha = \frac{|F_{Ey}|}{|F_{Ex}|}$$

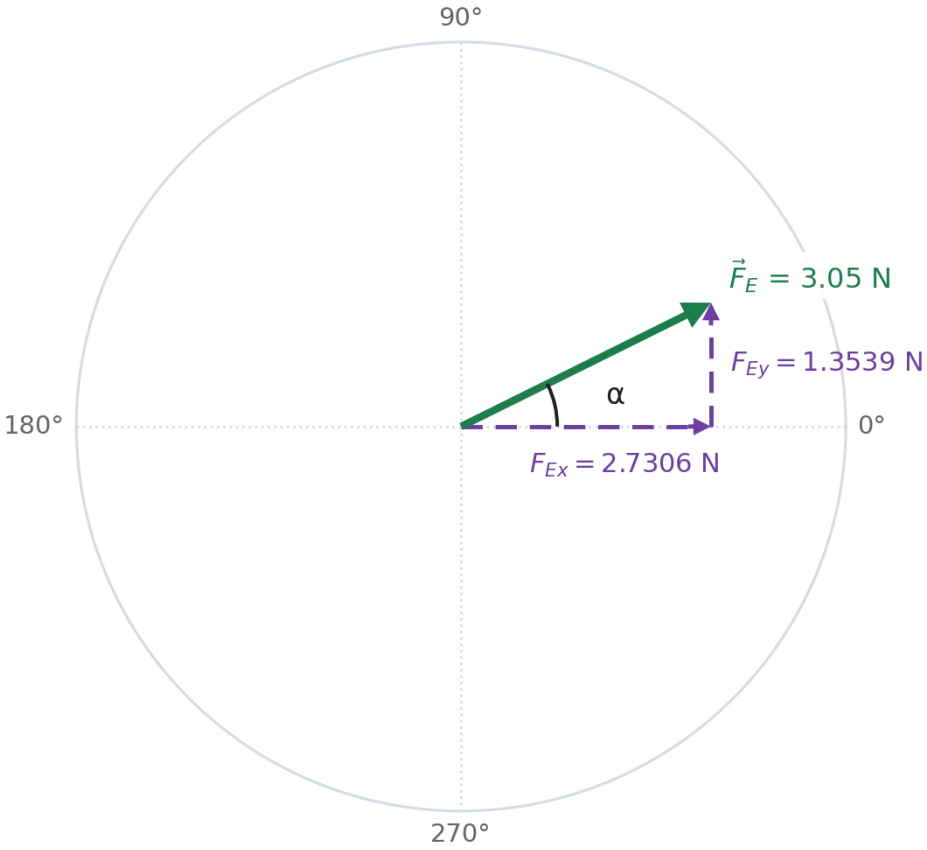
$$\alpha = \tan^{-1}\left(\frac{1.3539}{2.7306}\right) = 26.37°$$

Step 3 – Convert to the 360° scale

F_{Ex} is positive and F_{Ey} is positive, so F_E points right and up (between 0° and 90°). The 360° angle is just α:

$$\theta = 26.37°$$

Diagram: the equilibrant and its components



5. Test your answer at the link above.

Set the fourth string to 3.05 N at 26.4° and keep the three given forces. If the answer is right, the ring stays centred. Check by adding all four forces: each direction should sum to zero.

$$\Sigma F_x = -2.7306 + 2.7306 = 0 \qquad \Sigma F_y = -1.3539 + 1.3539 = 0$$

FINAL ANSWER

2. Components: see the table (each force's x and y parts)

3. F_{Ex} = 2.731 N, F_{Ey} = 1.354 N

4. $\vec{F}_E$ = 3.05 N [26.4°]

5. Force table setting: 3.05 N at 26.4°

SET 2

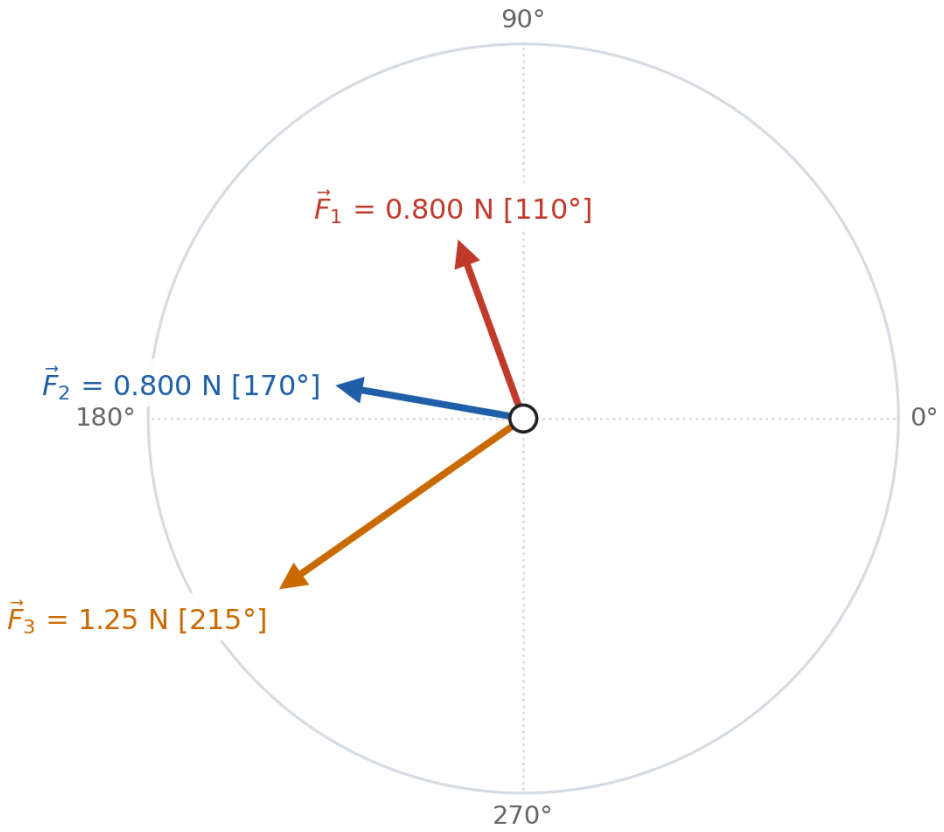
Forces to test

$F_1 = 0.800$ N [110°]
$F_2 = 0.800$ N [170°]
$F_3 = 1.25$ N [215°]

1. Draw a Free-Body diagram of the three forces given above.

Draw the ring as a dot. Each string pulls away from the ring at its angle on the 360° scale. Arrow lengths are proportional to the force sizes.

FBD of the ring (drawn to scale)



2. Calculate the magnitude of each component for the given forces.

Use the 360° angle of each force directly. A negative x component points left; a negative y component points down.

Formulas: $F_x = F\cos\theta$ $F_y = F\sin\theta$

$F_{1x} = (0.800)\cos 110° = -0.2736$ N

$F_{1y} = (0.800)\sin 110° = 0.7518$ N

$F_{2x} = (0.800)\cos 170° = -0.7878$ N

$F_{2y} = (0.800)\sin 170° = 0.1389$ N

$F_{3x} = (1.25)\cos 215° = -1.0239$ N

$F_{3y} = (1.25)\sin 215° = -0.7170$ N

Force	x component (N)	y component (N)
F_1 = 0.800 N [110°]	−0.2736	0.7518
F_2 = 0.800 N [170°]	−0.7878	0.1389
F_3 = 1.25 N [215°]	−1.0239	−0.7170
Sum	**−2.0854**	**0.1737**

3. Determine the components of the equilibrant using the net force in both the horizontal and vertical direction.

With the equilibrant added, the ring is in equilibrium, so each direction adds to zero:

Formula: $\Sigma F_x = 0$

Write out the x forces: $F_{1x} + F_{2x} + F_{3x} + F_{Ex} = 0$

Put in the sum: $-2.0854 + F_{Ex} = 0$

Add 2.0854 to both sides: $F_{Ex} = 2.0854 \text{ N}$

Formula: $\Sigma F_y = 0$

Write out the y forces: $F_{1y} + F_{2y} + F_{3y} + F_{Ey} = 0$

Put in the sum: $0.1737 + F_{Ey} = 0$

Subtract 0.1737 from both sides: $F_{Ey} = -0.1737 \text{ N}$

The equilibrant's components are the opposites of the sums.

4. Determine the direction of the equilibrant using the 360° scale so it will be easy to test on the force table.

Step 1 – Size of the equilibrant

F_{Ex} and F_{Ey} are at right angles, so use the Pythagorean theorem:

Formula: $F_E = \sqrt{F_{Ex}^2 + F_{Ey}^2}$

$F_E = \sqrt{(2.0854)^2 + (-0.1737)^2} = 2.0926 \text{ N}$

Step 2 – Angle α from the x-axis

α is the angle between F_E and the x-axis. The y part is opposite α and the x part is next to it (use sizes, no signs):

$$\text{Formula: } \tan\alpha = \frac{|F_{Ey}|}{|F_{Ex}|}$$

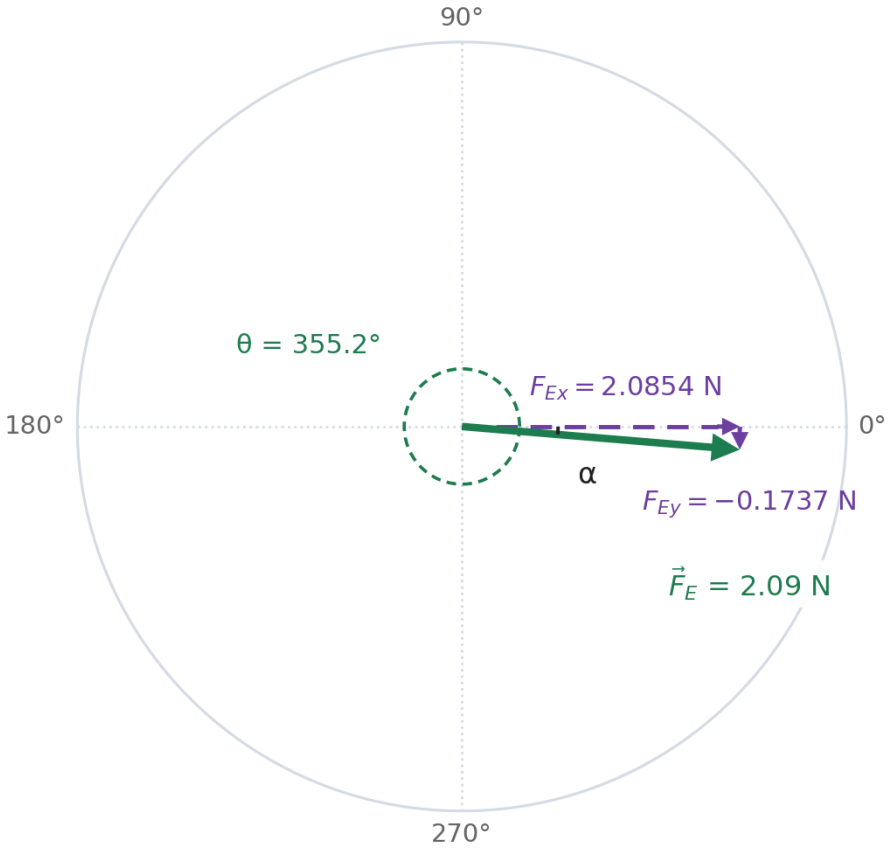
$$\alpha = \tan^{-1}\left(\frac{0.1737}{2.0854}\right) = 4.76°$$

Step 3 – Convert to the 360° scale

F_{Ex} is positive and F_{Ey} is negative, so F_E points right and down (between 270° and 360°). It is α below 0°, so subtract α from 360°:

$$\theta = 360° - 4.76° = 355.24°$$

Diagram: the equilibrant and its components



5. Test your answer at the link above.

Set the fourth string to 2.09 N at 355.2° and keep the three given forces. If the answer is right, the ring stays centred. Check by adding all four forces: each direction should sum to zero.

$$\Sigma F_x = -2.0854 + 2.0854 = 0 \qquad \Sigma F_y = 0.1737 + (-0.1737) = 0$$

FINAL ANSWER

2. Components: see the table (each force's x and y parts)

3. F_{Ex} = 2.085 N, F_{Ey} = −0.174 N

4. $\vec{F}_E$ = 2.09 N [355.2°]

5. Force table setting: 2.09 N at 355.2°

SET 3

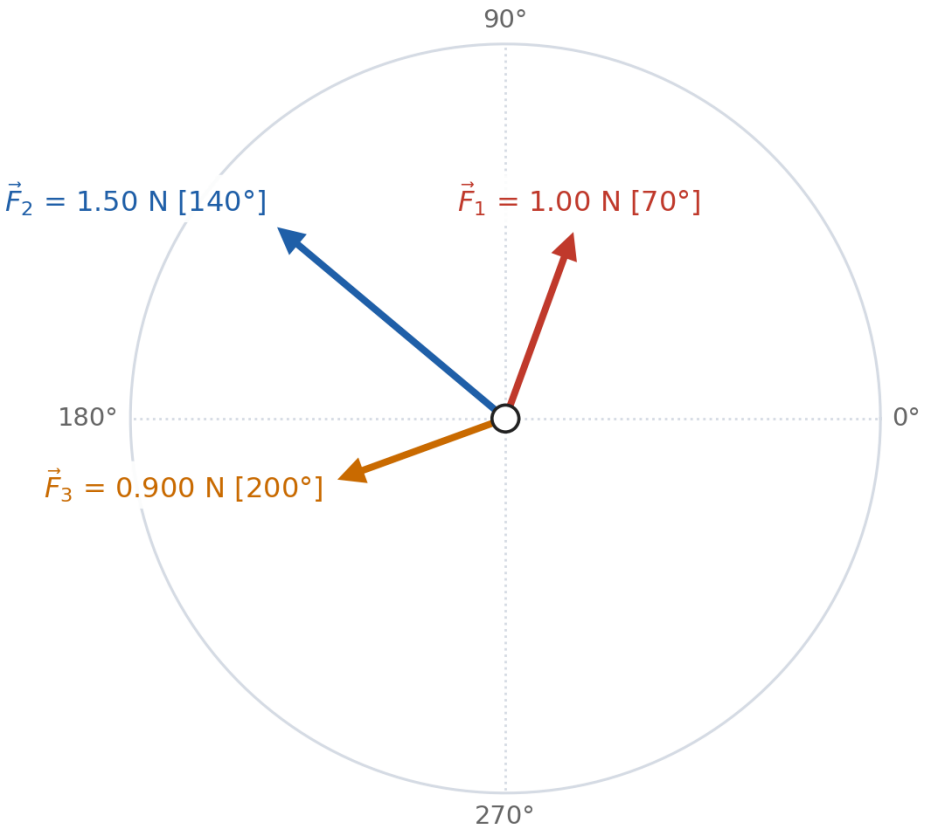
Forces to test

$F_1 = 1.00$ N [70°]
$F_2 = 1.50$ N [140°]
$F_3 = 0.900$ N [200°]

1. Draw a Free-Body diagram of the three forces given above.

Draw the ring as a dot. Each string pulls away from the ring at its angle on the 360° scale. Arrow lengths are proportional to the force sizes.

FBD of the ring (drawn to scale)



2. Calculate the magnitude of each component for the given forces.

Use the 360° angle of each force directly. A negative x component points left; a negative y component points down.

Formulas: $F_x = F\cos\theta$ $F_y = F\sin\theta$

$F_{1x} = (1.00)\cos 70° = 0.3420 \text{ N}$

$F_{1y} = (1.00)\sin 70° = 0.9397 \text{ N}$

$F_{2x} = (1.50)\cos 140° = -1.1491 \text{ N}$

$F_{2y} = (1.50)\sin 140° = 0.9642 \text{ N}$

$F_{3x} = (0.900)\cos 200° = -0.8457 \text{ N}$

$F_{3y} = (0.900)\sin 200° = -0.3078 \text{ N}$

Force	x component (N)	y component (N)
F_1 = 1.00 N [70°]	0.3420	0.9397
F_2 = 1.50 N [140°]	−1.1491	0.9642
F_3 = 0.900 N [200°]	−0.8457	−0.3078
Sum	**−1.6528**	**1.5961**

3. Determine the components of the equilibrant using the net force in both the horizontal and vertical direction.

With the equilibrant added, the ring is in equilibrium, so each direction adds to zero:

Formula: $\Sigma F_x = 0$

Write out the x forces: $F_{1x} + F_{2x} + F_{3x} + F_{Ex} = 0$

Put in the sum: $-1.6528 + F_{Ex} = 0$

Add 1.6528 to both sides: $F_{Ex} = 1.6528 \text{ N}$

Formula: $\Sigma F_y = 0$

Write out the y forces: $F_{1y} + F_{2y} + F_{3y} + F_{Ey} = 0$

Put in the sum: $1.5961 + F_{Ey} = 0$

Subtract 1.5961 from both sides: $F_{Ey} = -1.5961 \text{ N}$

The equilibrant's components are the opposites of the sums.

4. Determine the direction of the equilibrant using the 360° scale so it will be easy to test on the force table.

Step 1 – Size of the equilibrant

F_{Ex} and F_{Ey} are at right angles, so use the Pythagorean theorem:

Formula: $F_E = \sqrt{F_{Ex}^2 + F_{Ey}^2}$

$F_E = \sqrt{(1.6528)^2 + (-1.5961)^2} = 2.2976 \text{ N}$

Step 2 – Angle α from the x-axis

α is the angle between F_E and the x-axis. The y part is opposite α and the x part is next to it (use sizes, no signs):

$$\text{Formula: } \tan\alpha = \frac{|F_{Ey}|}{|F_{Ex}|}$$

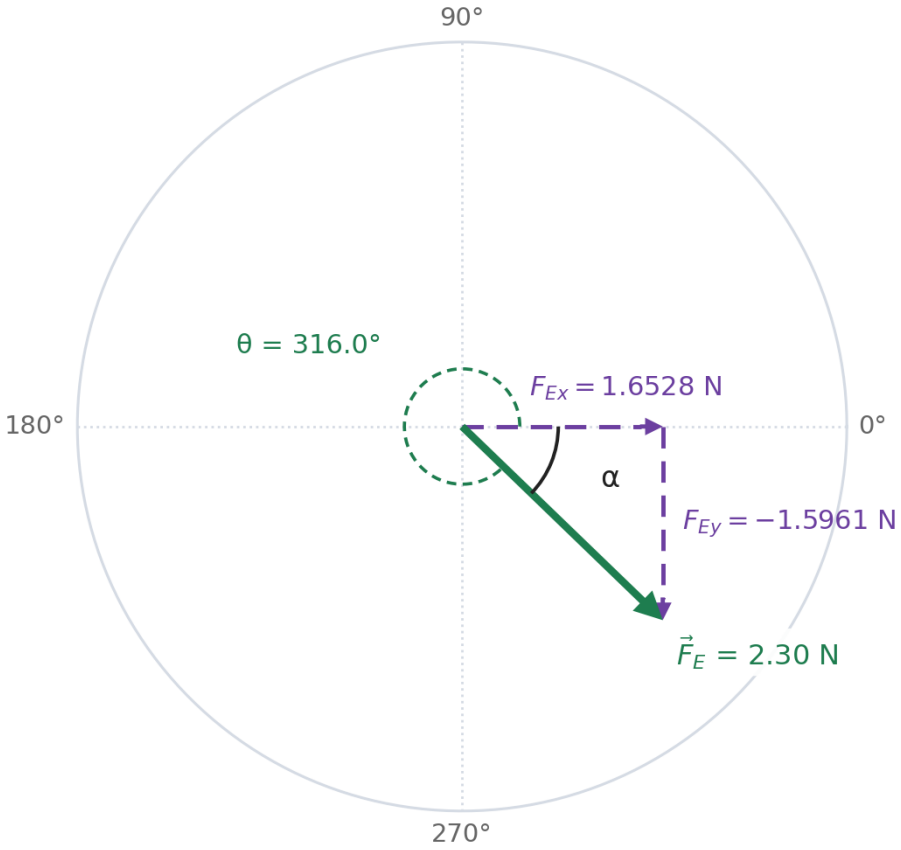
$$\alpha = \tan^{-1}\left(\frac{1.5961}{1.6528}\right) = 44.00°$$

Step 3 – Convert to the 360° scale

F_{Ex} is positive and F_{Ey} is negative, so F_E points right and down (between 270° and 360°). It is α below 0°, so subtract α from 360°:

$$\theta = 360° - 44.00° = 316.00°$$

Diagram: the equilibrant and its components



5. Test your answer at the link above.

Set the fourth string to 2.30 N at 316.0° and keep the three given forces. If the answer is right, the ring stays centred. Check by adding all four forces: each direction should sum to zero.

$$\Sigma F_x = -1.6528 + 1.6528 = 0 \qquad \Sigma F_y = 1.5961 + (-1.5961) = 0$$

FINAL ANSWER

2. Components: see the table (each force's x and y parts)

3. F_{Ex} = 1.653 N, F_{Ey} = −1.596 N

4. $\vec{F}_E$ = 2.30 N [316.0°]

5. Force table setting: 2.30 N at 316.0°